Lecture 6: Limit of Functions and Proving Limits Systematically

Mathematical Analysis

1 Limit of Functions

Definition 1.1 (Existence of a Limit at a Point). [def:limit-of-functions] Let (X, d_X) and (B, d_B) be metric spaces, let $A \subseteq X$ and $f : A \to B$, and let $c \in X$ be a limit point of A
If there exists a point $\ell \in B$ such that

$$\forall \varepsilon \in \mathbb{R}_+,\, \exists \delta \in \mathbb{R}_+,\, \forall x \in A,\, 0 < d_X(x, c) < \delta \implies d_B(f(x), \ell) < \varepsilon$$

then we say that f has a limit at c

Theorem 1.1 (Uniqueness of a Function Limit). [thm:function-limit-uniqueness] *With the setup of the existence definition, if a function f has a limit at a point c, then the limit is unique, ie*

$$\exists \ell : B,\, \forall \varepsilon \in \mathbb{R}_+,\, \exists \delta \in \mathbb{R}_+,\, \forall x \in A,\, 0 < d_X(x, c) < \delta \implies d_B(f(x), \ell) < \varepsilon$$

$$\iff$$

$$\exists! \ell : B,\, \forall \varepsilon \in \mathbb{R}_+,\, \exists \delta \in \mathbb{R}_+,\, \forall x \in A,\, 0 < d_X(x, c) < \delta \implies d_B(f(x), \ell) < \varepsilon$$

Proof. Assume that

$$\exists \ell_1,\, \forall \varepsilon \in \mathbb{R}_+,\, \exists \delta_1 \in \mathbb{R}_+,\, \forall x \in A,\, 0 < d_X(x, c) < \delta_1 \implies d_B(f(x), \ell_1) < \varepsilon$$

$$\exists \ell_2,\, \forall \varepsilon \in \mathbb{R}_+,\, \exists \delta_2 \in \mathbb{R}_+,\, \forall x \in A,\, 0 < d_X(x, c) < \delta_2 \implies d_B(f(x), \ell_2) < \varepsilon$$

where $\ell_1 \neq \ell_2$
Fix $\varepsilon > 0$ and choose the corresponding δ_1, δ_2. Since c is a limit point of A, choose $x \in A \cap B_{\min(\delta_1,\delta_2)}(c) \setminus \{c\}$
We know that

$$\forall x \in A \cap B_{\min(\delta_1,\delta_2)}(c) \setminus \{c\},\, d_B(f(x), \ell_1) + d_B(f(x), \ell_2) < 2\varepsilon$$

Then, according to triangle inequality,

$$d_B(\ell_1, \ell_2) \leq d_B(f(x), \ell_1) + d_B(f(x), \ell_2) < 2\varepsilon$$

According to epsilon criteria,

$$d_B(\ell_1, \ell_2) = 0$$

ie $\ell_1 = \ell_2$. Contradiction. So uniqueness is proven. □

Remark 1.1 (Board Comparison). [rem:sequence-function-limit-challenges] For a sequence $a : \mathbb{N}_1 \to X$ in a metric space (X, d) and a function $f : A \to B$, compare the two challenges

	Sequence	**Function**
Arbitrary tolerance	$\forall \varepsilon \in \mathbb{R}_+$	$\forall \varepsilon \in \mathbb{R}_+$
Restrictor	$\exists N \in \mathbb{N}_1$	$\exists \delta \in \mathbb{R}_+$
Input restriction	$n \geq N$	$0 < d_X(x, c) < \delta$
Output requirement	$d(a(n), \ell) < \varepsilon$	$d_B(f(x), \ell) < \varepsilon$

The output requirement must hold for every $n \in \mathbb{N}_1$ or $x \in A$ satisfying its input restriction

1. ε is arbitrary small
2. δ and N is used to restrict x st x is close enough to the limit
3. δ and N are all depends on ε

Convention 1.1 (The ε–δ Challenge)**.** [conv:epsilon-delta-challenge] A value $\ell \in B$ passes the ε–δ *challenge* for f at c if, for any proposed $\varepsilon > 0$, there exists a radius $\delta > 0$ such that for all $x \in A$ with $0 < d_X(x, c) < \delta$, $d_B(f(x), \ell) < \varepsilon$

Theorem 1.2 (Function Limit Challenge Replacement)**.** [thm:function-limit-challenge-replacement] *Let (X, d_X) and (B, d_B) be metric spaces, let $A \subseteq X$ and $f : A \to B$, let $c \in X$ be a limit point of A, and fix $\ell \in B$. Let g be a convergence challenge function. With balls taken in (X, d_X),*

$$\big(\forall \varepsilon \in \mathbb{R}_+,\, \exists \delta \in \mathbb{R}_+,\, \forall x \in A \cap B_\delta(c) \setminus \{c\}, \quad d_B(f(x), \ell) < \varepsilon\big)$$
$$\iff$$
$$\big(\forall \varepsilon \in \mathbb{R}_+,\, \exists \delta \in \mathbb{R}_+,\, \forall x \in A \cap B_\delta(c) \setminus \{c\}, \quad d_B(f(x), \ell) < g(\varepsilon)\big)$$

Proof. Similar to the sequence challenge-replacement proof □

2 Function Limit Operator

Definition 2.1 (Limit of a Function at a Point)**.** [def:function-limit-relation] Let (X, d_X) and (B, d_B) be metric spaces, let $A \subseteq X$ and $f : A \to B$, and let $c \in X$ be a limit point of A. We say that *f has limit $\ell \in B$ at c* if

$$\begin{gathered}\forall \varepsilon \in \mathbb{R}_+, \quad \exists \delta \in \mathbb{R}_+, \quad \forall x \in A, \\ 0 < d_X(x, c) < \delta \implies d_B(f(x), \ell) < \varepsilon\end{gathered}$$

Definition 2.2 (Limit Domain)**.** [def:function-limit-domain] A point in the limit domain if and only if

1. c is a limit point of the domain A
2. f has a limit at c

$$\operatorname{LimDom} : (A \to B) \to \mathcal{P}(X)$$
$$\operatorname{LimDom}(f) \equiv \left\{ c \in X \;\middle|\; \begin{gathered}\forall r \in \mathbb{R}_+, \quad B_r(c) \cap \big(A \setminus \{c\}\big) \neq \emptyset \\ \wedge \\ \exists \ell \in B, \quad \forall \varepsilon \in \mathbb{R}_+, \quad \exists \delta \in \mathbb{R}_+, \\ \forall x \in A, \quad 0 < d_X(x, c) < \delta \implies d_B(f(x), \ell) < \varepsilon\end{gathered} \right\}$$

`LaTeX: \LimDom (f)`

Proposition 2.1 (Well-Definedness of the Function Limit)**.** [prop:function-limit-well-defined] *With the setup of the limit relation, for every $f : A \to B$ and $c \in \operatorname{LimDom}(f)$, there exists a unique $\ell \in B$ such that f has limit ℓ at c*

Proof. Existence follows from the definition of the limit domain. The limit is unique by the uniqueness theorem □

Definition 2.3 (Function Limit Operator)**.** [def:function-limit-op]

$$\lim : \Pi f : (A \to B), \quad \operatorname{LimDom}(f) \to B$$
$$\lim(f)(c) = \ell \iff \begin{gathered}\forall \varepsilon \in \mathbb{R}_+, \quad \exists \delta \in \mathbb{R}_+, \quad \forall x \in A, \\ 0 < d_X(x, c) < \delta \implies d_B(f(x), \ell) < \varepsilon\end{gathered}$$

`LaTeX: \fnlim (f)(c)`

Convention 2.1 (Function Limit Notation). [conv:function-limit-notation] For the function limit operator, write

$$\lim_c(f) \equiv \lim(f)(c)$$

LaTeX: \fnlim _c(f)

3 Properties of Function Limits

Theorem 3.1. *If we can propose a convergence sequence a which codomain is in the domain of a function f exculding its c, where $\lim a = c$, then the limit of the function f at c is equal to the limit of the sequence $n \mapsto f(a(n))$.*

$$f : A \to B,\ \lim_c(f) = \ell \iff \forall a : \mathbb{N} \to A \setminus \{c\},\ \lim a = c,\ \lim(n \mapsto f(a(n))) = \ell$$

4 One-Sided Limits

Definition 4.1 (Right-Hand Limit). [def:right-hand-limit] Let $A \subseteq \mathbb{R}$, let (B, d_B) be a metric space, and let $f : A \to B$. Let $c \in \mathbb{R}$ satisfy

$$\forall r \in \mathbb{R}_+, \quad \exists x \in A, \quad 0 < x - c < r$$

We say that f has *right-hand limit* $\ell \in B$ at c if

$$\forall \varepsilon \in \mathbb{R}_+,\ \exists \delta \in \mathbb{R}_+,\ \forall x \in A \cap (c, c+\delta),\ d_B(f(x), \ell) < \varepsilon$$

Definition 4.2 (Left-Hand Limit). [def:left-hand-limit] Let $A \subseteq \mathbb{R}$, let (B, d_B) be a metric space, and let $f : A \to B$. Let $c \in \mathbb{R}$ satisfy

$$\forall r \in \mathbb{R}_+, \quad \exists x \in A, \quad 0 < c - x < r$$

We say that f has *left-hand limit* $\ell \in B$ at c if

$$\forall \varepsilon \in \mathbb{R}_+,\ \exists \delta \in \mathbb{R}_-,\ \forall x \in A \cap (c+\delta, c),\ d_B(f(x), \ell) < \varepsilon$$

Theorem 4.1 (Uniqueness of One-Sided Limits). [thm:one-sided-limit-uniqueness] *Under the assumptions of the right-hand or left-hand definition, the corresponding one-sided limit, if it exists, is unique*

Proof. The argument is the same as for two-sided limits, choosing x in the shared one-sided interval using the corresponding accumulation assumption □

Definition 4.3 (Right-Hand Limit Domain). [def:right-hand-limit-domain] Let $A \subseteq \mathbb{R}$ and let (B, d_B) be a metric space

$$\mathrm{LimDom}_+ : (A \to B) \to \mathcal{P}(\mathbb{R})$$

$$\mathrm{LimDom}_+(f) \equiv \left\{ c \in \mathbb{R} \;\middle|\; \begin{array}{c} \forall r \in \mathbb{R}_+, \quad A \cap (c, c+r) \neq \emptyset \\ \wedge \\ \exists \ell \in B, \quad \forall \varepsilon \in \mathbb{R}_+, \quad \exists \delta \in \mathbb{R}_+, \\ \forall x \in A \cap (c, c+\delta), \quad d_B(f(x), \ell) < \varepsilon \end{array} \right\}$$

LaTeX: \LimDomRight (f)

Definition 4.4 (Left-Hand Limit Domain). [def:left-hand-limit-domain] Let $A \subseteq \mathbb{R}$ and let (B, d_B) be a metric space

$$\mathrm{LimDom}_- : (A \to B) \to \mathcal{P}(\mathbb{R})$$

$$\mathrm{LimDom}_-(f) \equiv \left\{ c \in \mathbb{R} \;\middle|\; \begin{array}{c} \forall r \in \mathbb{R}_+, \quad A \cap (c-r, c) \neq \emptyset \\ \wedge \\ \exists \ell \in B, \quad \forall \varepsilon \in \mathbb{R}_+, \quad \exists \delta \in \mathbb{R}_-, \\ \forall x \in A \cap (c+\delta, c), \quad d_B(f(x), \ell) < \varepsilon \end{array} \right\}$$

`LaTeX: \LimDomLeft (f)`

Proposition 4.1 (Well-Definedness of One-Sided Limits). [prop:one-sided-limit-well-defined] *For every $f : A \to B$ and $c \in \mathrm{LimDom}_+(f)$, there is a unique right-hand limit in B. The same holds for $c \in \mathrm{LimDom}_-(f)$ and the left-hand limit*

Proof. Existence follows from the definitions of the right-hand and left-hand limit domains. Uniqueness follows from the one-sided uniqueness theorem □

Definition 4.5 (Right-Hand Limit Operator). [def:right-hand-limit-op] The operator

$$\lim_+ : \Pi f : (A \to B), \quad \mathrm{LimDom}_+(f) \to B$$

sends f and $c \in \mathrm{LimDom}_+(f)$ to the unique right-hand limit of f at c

`LaTeX: \rightfnlim (f)(c)`

Definition 4.6 (Left-Hand Limit Operator). [def:left-hand-limit-op] The operator

$$\lim_- : \Pi f : (A \to B), \quad \mathrm{LimDom}_-(f) \to B$$

sends f and $c \in \mathrm{LimDom}_-(f)$ to the unique left-hand limit of f at c

`LaTeX: \leftfnlim (f)(c)`

Convention 4.1 (One-Sided Limit Notation). [conv:one-sided-limit-notation] For $c \in \mathrm{LimDom}_+(f)$ or $c \in \mathrm{LimDom}_-(f)$, respectively, write

$$\lim_{c^+}(f) \equiv \lim_+(f)(c), \qquad \lim_{c^-}(f) \equiv \lim_-(f)(c)$$

`LaTeX: \poslim {3}(f), \neglim {3}(f)`

Theorem 4.2 (Two-Sided and One-Sided Limits). [thm:two-sided-one-sided-limits] ***TODO:*** *State when a two-sided limit is equivalent to equal right-hand and left-hand limits, including the required domain assumptions*

Proof. **TODO:** Prove both directions □

Remark 4.1 (Error Bounds for One-Sided Limits). [rem:one-sided-error-bounds] **TODO:** Adapt the shared error-bound method to the right-hand and left-hand input regions

5 Limits at Infinity

Definition 5.1 (Limit at Positive Infinity). [def:limit-at-positive-infinity] **TODO:** Assume the real domain is unbounded above and define a finite output limit as the input tends to positive infinity

Definition 5.2 (Limit at Negative Infinity). [def:limit-at-negative-infinity] **TODO:** Assume the real domain is unbounded below and define a finite output limit as the input tends to negative infinity

Convention 5.1 (Limit Notation at Infinity). [conv:limit-at-infinity-notation] **TODO:** Introduce the notation for limits as the input tends to positive or negative infinity

Theorem 5.1 (Uniqueness of Limits at Infinity). [thm:limit-at-infinity-uniqueness] ***TODO:*** *State uniqueness under the corresponding unbounded-domain assumption*

Proof. **TODO:** Adapt the uniqueness argument using the corresponding input tail □

Remark 5.1 (Error Bounds for Limits at Infinity). [rem:infinity-error-bounds] **TODO:** Adapt the shared error-bound method using a real input threshold M(t)

6 A Systematic Method for Proving Limits

The shared error-bound method covers finite-value limits of sequences, two-sided function limits, one-sided limits, and limits as the input tends to infinity. The input restriction changes with the approach, while the output tolerance is handled using challenge functions. Infinite output limits use arbitrarily large output thresholds instead

Remark 6.1 (Shared Proof Method). [rem:systematic-limit-method] **TODO:** Summarize the shared procedure: choose a candidate limit, derive an error bound, verify the challenge function, and supply the corresponding radius or threshold

Remark 6.2 (Using the Ball Radius as the Parameter). [rem:ball-radius-parameter] In the challenge-function method, when the input-radius parameter itself supplies the witness δ, the following condition suffices

$$\forall \varepsilon \in \mathbb{R}_+, \quad \forall x \in A \cap B_\varepsilon(c) \setminus \{c\}, \quad d_B(f(x), \ell) \leq g(\varepsilon)$$

Here ε is the input-radius parameter, and $g(\varepsilon)$ is the output-error bound for a challenge function g. A local restriction such as an input radius at most 1 can instead be handled by choosing $\delta = \min(\varepsilon, 1)$ and verifying the output bound on that smaller ball